

Prelab 3.1 IoT Data Collection 3 - Industry data communication protocols: Power meter (Modbus) and vibration (IO-Link)

Learning Goals

Students will be able to:

1. Define and describe Modbus protocol
2. Define and describe IO-Link protocol
3. Explain TCP/IP and REST API

1.1 Introduction

Seamless real-time communication is the core prerequisite for all Industry 4.0 and Industrial Internet of Things (IIoT) implementations. In Lab3, we will collect data from two IoT devices, a power meter ([Acuvim L series](#), AccuEnergy, Canada), and a vibration sensor ([VVB001](#), IFM, Germany) with a IO-Link master ([AL1302](#), IFM, Germany). Please visit and look into both web pages of the devices. For the power meter, Modbus protocol will be used. On the other hand, for the vibration sensor, the IO-Link protocol will be used. When we read data from the vibration sensor through the IO-Link master, we use REST API via TCP/IP (so-called EtherNet/IP) communication. But the communication between the vibration sensor and the IO-Link master is being made by IO-Link protocol. This prelab is to install a related package on Raspberry Pi and to be familiar with the industry standard protocols. Modbus and IO-Link are representative protocols in Fieldbus communications. And TCP/IP is a representative protocol in Industrial Ethernet communications.

On top of that, it is highly recommended to go through JSON data format and how to parse it from online resources.

1.2 Python package installation on Raspberry Pi

'pymodbus' package installation

Install 'mymodbus' package on your Raspberry Pi as below.

 **Raspberry Pi - Terminal**

```
pip3 install pymodbus==2.5.3
```

After installing 'pymodbus' package, see if it is installed successfully using the command 'pip3 list'.

Task 1.1

Capture your terminal window as Figure 1 after running 'pip3 list'.

```
pi@TA: ~  
File Edit Tabs Help  
pip 18.1  
psutil 5.5.1  
pycairo 1.16.2  
pycrypto 2.6.1  
pycups 1.9.73  
pyftdi 0.53.3  
pygame 1.9.4.post1  
Pygments 2.3.1  
PyGObject 3.30.4  
pyinotify 0.9.6  
PyJWT 1.7.0  
pylint 2.2.2  
pymodbus 2.5.3  
pyOpenSSL 19.0.0  
pyparsing 3.0.6  
pyserial 3.5  
pysmbc 1.0.15.6  
python-apt 1.8.4.3  
python-dateutil 2.8.2  
pyusb 1.2.1  
pyxdg 0.25  
rainbowhat 0.1.0  
reportlab 3.5.13  
requests 2.21.0
```

Figure 1 Checking the installed Python packages

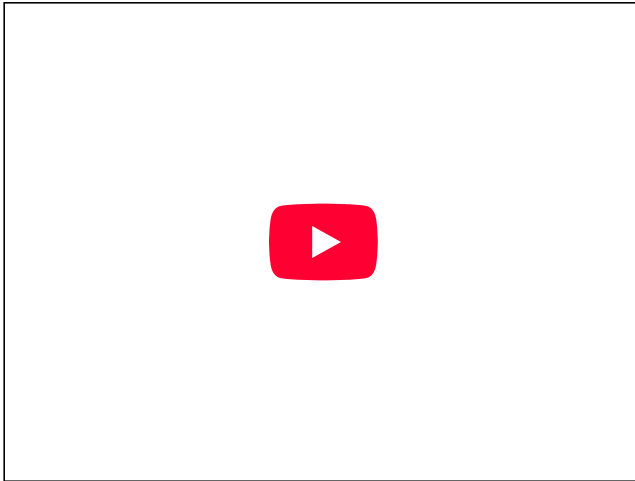
```
ssh - ME-597-llot - Visual Studio Code  
adafruit-circuitpython-adxl34x 1.12.18  
adafruit-circuitpython-busdevice 5.2.15  
adafruit-circuitpython-connectionmanager 3.1.6  
adafruit-circuitpython-requests 4.1.15  
adafruit-circuitpython-typing 1.12.3  
Adafruit-PlatformDetect 3.87.0  
Adafruit-PureIO 1.1.11  
binho-host-adapter 0.1.6  
contourpy 1.3.3  
cyclor 0.12.1  
fonttools 4.61.1  
kiwisolver 1.4.9  
matplotlib 3.10.8  
numpy 2.4.2  
packaging 26.0  
pillow 12.1.0  
pip 25.1.1  
pyftdi 0.57.1  
pymodbus 2.5.3  
pyparsing 3.3.2  
pyserial 3.5  
python-dateutil 2.9.0.post0  
pyusb 1.3.1  
RPi.GPIO 0.7.1  
rpi_ws281x 5.0.0  
scipy 1.17.0  
six 1.17.0  
smbus 1.1.post2  
sysv_ipc 1.2.0  
typing_extensions 4.15.0  
(lab-venv) pi@robertclaud:~/Documents/lab $
```

Task 1.2

Address the following questions by checking provided resources.

```
In [5]: from IPython.display import YouTubeVideo  
        YouTubeVideo('txi2p5_0jKU')  
  
#Watch this video to answer question 1.
```

Out[5]:



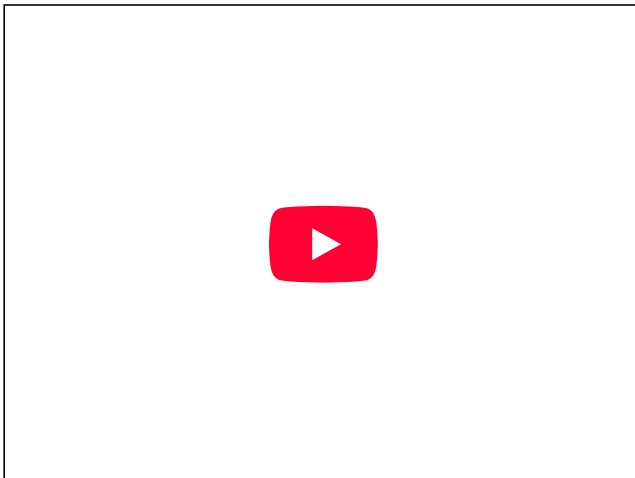
1) Explain what is Modbus protocol?

Modbus is the oldest open automation communication protocol for industrial devices. It is most commonly used to facilitate Supervisory Control and Data Acquisition or SCADA. It is used as a common communication protocol for devices working together in a system even if they are made by different manufacturers. Modbus is designed based on a master-slave architecture. This means that a single master computer sends messages to its slaves, commonly sensors or other peripheral devices, and expects responses of either sensor readings or acknowledgement.

```
In [2]: from IPython.display import YouTubeVideo
YouTubeVideo('YDbWt_wE3YE')

#Watch this video to answer question 2.
```

Out[2]:



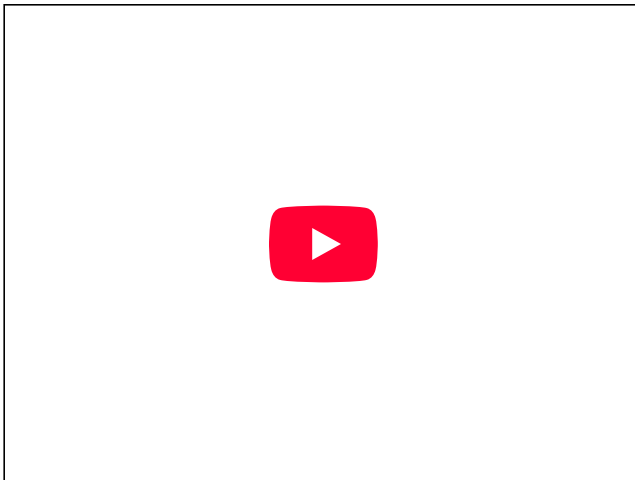
2) Explain what is IO-Link protocol?

The IO-Link protocol is a protocol that connects sensors and actuators to an interface with PROFINET. This PROFINET can communicate with a centralized controller that can display visual data or communicate with higher level SCADA or MES devices. It allows for easy install, faster data processing, and data parameterization. IO-Link exists at a lower level than MODBUS since IO-link handles binary and analog inputs and outputs from sensors and actuators rather than sophisticated messages.

```
In [3]: from IPython.display import YouTubeVideo
        YouTubeVideo('614QGgw_FA4')

        #Watch this video to answer question 3.
```

Out[3]:



3) Explain what is TCP/IP protocol?

TCP/IP is an internet communication protocol that describes how messages should be sent and recieved over the internet. It deccribes how they should be

1. Broken Down
2. Addressed
3. Transmitted
4. Routed
5. Received

TCP defines how applications should create communication channels and manages how messages should be broken down and reassembled. IP defines how to address devices to ensure that messages are recieved by the correct device.

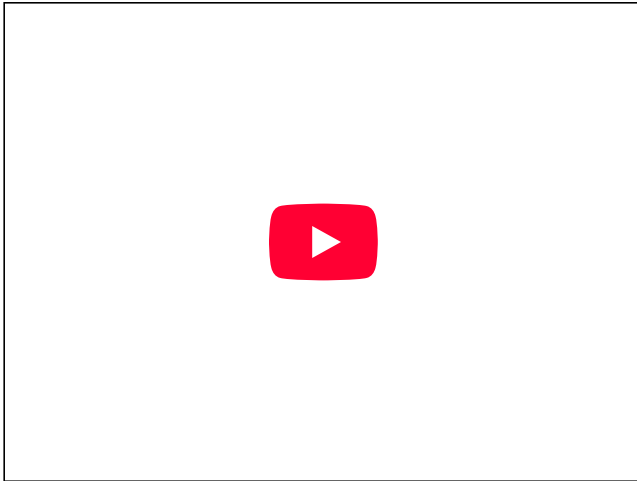
TCP/IP has four layers:

1. Application Layer: Standardized data exchange protocols
 2. Transport Layer: End-to-end communication accross network using TCP
 3. Network Layer: Connects networks for messages to be sent and received outside of the network boundaries.
 4. Physical Layer: Messages that only operate on a link or nodes and devices that exist on a network.
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```
In [4]: from IPython.display import YouTubeVideo
        YouTubeVideo('1sMQRaeKNDk')

        #Watch this video to answer question 4.
```

Out[4]:



4) Explain what is REST API?

REST API stands for Representational State Transfer API. It is a communication protocol for web applications to communicate with a cloud or server. Some of its benefits include its stability, statelessness, simplicity, scalability, and its high performance due to caching.

REST API consists of a request from an application and the response received from the server. Requests send commands for CRUD or Create, Read, Update, and Delete. Over RESTful communication, these translate into the POST, GET, PUT and DELETE operations. Requests consist of headers, operation, endpoint to use, and the request body. The response typically is a simple json message with information from the cloud or server.

5) What can be the applications of power and vibration monitoring in the industry?

Some examples of power and vibration monitoring in industry include:

1. Monitoring the power output of a generator.
 2. Ensuring that large spinning equipment is not exceeding its manufacturing specification.
 3. Identifying where electrical faults are caused for easier diagnosis and maintenance.
 4. Giving live feedback data in automotive applications for battery health and the degradation of spinning components.
 5. Identifying when motors are physically blocked from rotating based on their electrical resistance.
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Please continue to [Prelab 3.2 here](#).